

Case Study 1: Tire Road Wear Particles TRWP

Problem:

The tyre production worldwide is about 17 million tons of tires/year with China contributing to most of them followed by Asia and then Europe and America. Apart from the production of tires disposing them is an equally challenging task and it involves:

- 2% air deposition
- 18% captured by wastewater systems
- 61% deposited into soil
- 16% sedimentation in freshwater
- 2-5% reaching estuaries via rivers

The contribution of TRWP into the environment as microplastic pollution is around 15% and is emitted due to the friction between the vehicle types (rubber – density of roughly around 1) and the road surface as it wears away the surface of the tyre and the road.

In conventional tyres there is little to no abrasion patterns (smearing) however in high performance vehicles there is a surface abrasion pattern that is formed. There is a mixture of tyre tread material and pavement material in equal parts (50-50) that are usually the shape of cigars below 1mm in length and typically 70-100 micrometres and have an average density of 1.8 Mg/m^2 . They look like tiny, elongated particles and studies show that most of these particles are deposited on the roadside (around 50%) with 2% being inhaled. A small fraction of it is released into the air. Research suggests that it is unlikely to cause risk to human health and adversely affect fresh water we must still consider long term effects and the presence, fate and transport of TRWP in air, soil rivers and oceans.

Around 15% of the macro plastics (to become microplastics) are used in tyre production which is bad due to their ability to enter the human body and cause inflammation, respiratory issues and other serious health problems. The gap between the tyre microplastic contribution per year varies from 10% all the way to 85% which is a big problem and needs to be dealt with. There are 17 million tonnes of tires made per year with 3.4 million tons of total emissions per year suggesting that each tyre has approximately 20% fo the emission contribution.

Main aim:

- 1) Remove toxic materials such as zinc oxide, PAHs, and 6PPD where possible, or replace them with safer alternatives to reduce harm to human health and the environment.
- 2) Reduce TRWP generation by designing tyres that favour smear wear over abrasive wear, which produces fewer and smaller particles while forming a protective oxidised layer.
- 3) Use high-toughness compounds that slow down wear rates, decreasing both the volume and size of emitted particles and enhancing tyre durability.
- 4) Maintain performance balance by optimising the trade-off between low rolling resistance (Rolling resistance is the amount of energy a tyre uses over a defined distance to overcome the resistance and is essentially the energy lost as the tire deforms and moves) and good wet grip, ensuring both energy efficiency and safety.
- 5) Explore future solutions like filtering, capturing, or recycling TRWPs as part of an extended approach to minimise environmental pollution.

Requirement while making a tire

- The tire must be able to grip the road even in wet conditions which is critical for the safety performance and comfort of driving.
- It must be wear resistant as it must be able to resist abrasion, fatigue and cutting to ensure long life and reduce maintenance.
- It must also have a rolling resistance improving fuel efficiency and lowering CO2 emissions.
- It must be thermally stable while being processable.
- High toughness and ideally smear wear.

Obvious no choices and Composition

Tyres are composed of various materials, including zinc, sulfur, silicon and chemical additives like 6PPD which are both harmful to the environment and human health. They contribute to the TRWP that are released into the environment as a tyre wears down.

- Mineral content makes up 53% of Road Particles (RP), 61% of TRWP, and 16% of Tyre Particles (TP).
- These particles contain toxic substances that may:
 - Bind to seaweed and enter marine food chains
 - Affect pregnant women and fetal development
 - Leach in low pH (acidic) environments, potentially increasing cancer risk.
- Toxic substances:
 - Zinc oxide: Harmful to aquatic life and can be bioaccumulate
 - PAHs (Polycyclic Aromatic Hydrocarbons): Found in both tyres and road materials and are linked to cancer and environmental toxicity.
 - 6PPD which is a common antioxidant in tyres while relatively safe on its own it reacts with ozone or oxygen to form 6PPD-quinone, which is highly toxic particularly to fish.
- **Health Effects**
 - Studies show that TRWP does not cause cell death or oxidative stress, but do trigger inflammation (increased TNF- α , a proinflammatory marker) in a dose-dependent manner.
 - 6PPD-quinone once formed, can accumulate in fish leading to their death and raising concerns for broader ecological impacts.
- PM2.5 stands for Particulate Matter that is 2.5 micrometres or smaller in diameter are tiny particles in the air that can be made from dust, dirt, soot or smoke. It is harmful as it can cause respiratory issues. We can infer from this that particles with smaller diameters have a larger count of inhalation. Furthermore, due to their small diameters their mass is negligible, and they can be leached (washed by water into the surrounding environment). Hence the toxicity level increases.

Tire Wear is proportional to the friction energy where:

$\text{Wear} \cong \text{Friction Energy} = \text{Friction Force} \times \text{Sliding distance}$

$\text{Friction Force} = \text{Mass}_{\text{vehicle}} \times \text{Acceleration of vehicle}_{\text{longitudinal and lateral}}$

Things about a car that can affect a tyre's wear

- Tyre characteristics: Size, tread depth, construction, tyre pressure and temperature, contact patch area, chemical composition, accumulated mileage.
- Road surface characteristics: Pavement construction, aggregate rocks, binder (bitumen, cement), porous asphalt, macro and micro texture, porosity, condition, road surface wetness, road dust loading in surface texture.
- Driving behaviour/Vehicle operation: Speed, linear and radial acceleration, frequency and extent of braking and cornering.
- Vehicle characteristics: vehicle weight and distribution of loads, location of driving wheels, wheel alignment, engine power, power/unassisted steering, electronic braking systems, suspension type and condition.
- Tyre design and Road surface as not very high weighing factors that influence the amount of wear experienced by a tyre compared to the driving style which counts a lot more. For example tyres from high performance cars like F1 wear out much faster than a low performance car and thus will also have a very different wear pattern.
 - High performance – abrasive wear
 - Low performance – smear wear

Difference in the type of wear

- Abrasion refers to the process of wearing away a material's surface through friction caused by the mechanical rolling or sliding of solids and liquid over it. When tyres wear down due to contact with the road surface, two primary types occur in practice. These differ in how the rubber degrades and the nature of the particles that are released into the atmosphere.
 - **Abrasive wear:** involves the formation of large at evenly spaced intervals in the tyre tread surface that grow over time leading to chunks of rubber tearing off in relatively large pieces. The crack formation is macro-scale which results in large rubber debris or particles being released and these chunks tend to accumulate at the roadside or the near environment. These bits are relatively easier to clean up or control and it mainly happens due to the heat and friction working together that wear down the tyre. This results in short lasting tyres and the debris generated is relatively less harmful to the environment as the chunks are larger and can float on water. Usual in high friction aggressive driving conditions. These larger debris pieces are also less likely to be inhaled or ingested and therefore pose a lower immediate health risk compared to finer particles.
 - **Smear wear:** characterised by the formation of many small microcracks occurring at a much higher frequency across the rubber surface. It produces small fine particles (microplastics) which are larger in quantity compared to abrasive wear. These particles stick onto the surface forming an oxidised protective layer on the tyre surface that protects the bulk rubber from any further excessive crack propagation and further wear. It happens due to the chemical reactions during friction (tribo-chemical degradation which is an oxidised sticky layer that protects the bulk rubber surface from excessive crack initiation and propagation) and this wear type helps tyres last longer which is very common in regular vehicles. The smaller particles produced are of larger concern as they become airborne or get washed into waterways which raise concerns about microplastic pollution and inhalation risks. These airborne fine particles have also been found in the lungs of humans and may accumulate in aquatic species through ingestion, leading to long-term environmental and health concerns. Usual in low friction conditions. Moreover, tribo-

chemical degradation may lead to the release of reactive chemical compounds such as 6PPD-quinone, which forms when 6PPD (an antioxidant used in tyres) reacts with ozone. This compound is known to be highly toxic to aquatic life. So, although smear wear produces less visible debris and increases tyre life, the fine, volatile particles it releases may pose greater long-term ecological and human health risks than the chunkier debris from abrasive wear.

Testing Wear Only

Our main objective now is to understand how rubber (specifically, tread compounds from tires) respond to dynamic abrasion by simulating real world wear. They are mainly used to understand the frictional force, friction energy and size of wear debris.

- 1) Analyse the rubber properties by testing the viscoelastic behaviour of the tread compound and how it stretches and returns.
- 2) Cut the rubber samples for the tires and mount them in a device with a blade, actuator and load cell
- 3) Perform uni-directional dynamic abrasion test – apply a vertical force with a blade and move the blade in a rotating motion to simulate abrasion. Observe and analyse crack formation and wear on the rubber.

At low energy severity it is smearing and at high energy severity it is abrasive.

Solution

To effectively reduce TRWP while maintaining tyre performance the tread compound design must balance several competing properties: low rolling resistance, high wet grip and wear resistance.

1. Material selection and Toxicity Reduction
 - Natural Rubber (NR) is harvested from tree sap and contains natural proteins and chemicals that make it tougher, providing low rolling resistance, excellent tear resistance, and flexibility. However, it oxidizes easily at higher temperatures due to its unsaturated polymer structure.
 - Isoprene Rubber is the synthetic version of natural rubber but lacks the natural proteins, making it generally less tough than NR.
 - Styrene-Butadiene Rubber (SBR) is commonly used in passenger car tires for its superior wet grip due to high viscoelasticity, but it requires fillers like carbon black or silica to reinforce its strength and durability.
 - Butadiene Rubber (BR) is used in tire treads for its flexibility and fatigue resistance, often blended with NR and SBR to balance properties.
 - Different tire types use these rubbers variably: passenger cars mainly use SBR, trucks and buses mostly use NR, and BR is found in all tire types.
 - Tires contain additives like zinc oxide, polycyclic aromatic hydrocarbons (PAHs), and 6PPD; 6PPD can transform into toxic 6PPD-quinone when exposed to oxygen or ozone, posing environmental and health risks, especially to aquatic life.

2. Fillers and Bi-Continuous Phase Structure

- Carbon Black: Created by controlled soot particles, it reinforces rubber, improves wear resistance, and gives tires their black colour. Essential for SBR but NR can survive without it.
- Silica: Preferred filler over the past 30 years for improving wet grip and optimizing rolling resistance by affecting viscoelastic properties.
- Filler Properties: Size, shape, and surface chemistry affect interaction with rubber; smaller particles with higher surface area improve wear resistance and grip.

3. Performance Trade-offs

- Rolling Resistance: Requires low viscoelastic energy dissipation (soft, low modulus materials like NR), which improves fuel efficiency but compromises grip and wear.
- Wet Grip: Requires high viscoelastic energy dissipation (like SBR with silica fillers) for better safety but increases rolling resistance.
- Wear Resistance: Needs high toughness, high energy dissipation materials that resist abrasion, often conflicting with rolling resistance goals.

4. Material structure and Compound and Design

- Bi-continuous Structure: Combines unfilled NR phase for low rolling resistance and crack resistance with highly filled SBR phase for wet grip and wear resistance.
- End-Chain Modification: SBR chains are chemically modified to attract fillers, ensuring fillers concentrate in the SBR phase, optimizing performance.
- Crosslinking: Higher crosslink density strengthens the material; different crosslink types affect degradation behaviour under friction.
- Carbon Black Content: More carbon black increases toughness and wear resistance; certain grades (CBS, CBH) are better under high stress.

5. Testing and Characterization

- Dynamic Mechanical Analysis (DMA): Measures energy dissipation (tan delta) at various temperatures and frequencies to simulate rolling resistance (low frequency, ~10 Hz) and wet grip (high frequency, ~100-1000 Hz).
- Superposition Principle: Allows correlation of material behaviour at different temperatures and frequencies, aiding in optimizing tire compounds.

6. Processing

- Low Temperature Curing: Produces uniform crosslinked networks with low energy loss and higher toughness.
- Filler-Rubber Interaction: Achieved via chain-end modification to ensure fillers adhere well to the polymer matrix.

How is a tyre made

How are tyres made?

Mixing and Extruding Process:

They make the rubber in sheets. They have a big mixing machine that basically mixes it all up and make it into a dough and then pump it out and line it up into sheets like this and then they cut the sheet into certain sizes and lengths and then start going through the tyre assembly process where they build the carcass of a tyre. Then they put the various bits that are built into the tyre. So, the tyres are laid out as a big fat cylinder to start with, and they build it all up and then they inflate it to make it into the shape of a tyre by pushing the middle of the tyre out. They have rubber diaphragm's that do it and then they create a structure that's got all the various bits of a normal radial tyre. Nearly all passenger car tyres, and truck tyres are made using the radial technology.

Assembling process:

Assembling, inflating and curing the tyre. You make this thing called a green tyre, you put it into a hot press, that hot press is then put under pressure and you end up with a tyre.

- 1) Assembly: Layers (inner liner, plies, beads, belts, sidewall, tread) are built on a drum to make a soft green tyre.
- 2) Inflation: The green tyre is placed in a Mold, and a bladder inside is inflated with steam to shape the tyre.
- 3) Curing: Heat (~150–180°C) vulcanizes the rubber, hardening it and forming the final tread pattern from the Mold.

Key point

Highly filled SBR phase can be achieved by the functionalized SBR with Chain end modification. Bi-continuous structure optimises two extreme trading off performances. We make bi continuous structures by shearing. Low temperature cure gives uniform crosslinked network which gives lower tanD and higher toughness. No coating. We want more contact area material which is softer and has a low Young's modulus. Less loading filler gives better energy loss and saturation of wear. Lower wear resistance: Less reinforcement means the tyre wears out faster. Lower energy loss: The tyre is more elastic and returns more energy, improving fuel efficiency. Saturation of wear: At lower filler levels, the wear rate gets worse quickly but eventually reaches a steady rate (saturation point).

Solution Summary

To reduce tyre wear particles (TRWP) while maintaining tyre performance, the ideal approach is to design a high-toughness tread compound that favors smear wear (producing smaller, less harmful particles) rather than abrasive wear. This involves using a bi-continuous structure combining unfilled natural rubber (NR) for low rolling resistance and crack resistance, with highly filled styrene-butadiene rubber (SBR) for better wet grip and wear resistance. The fillers (carbon black and silica) should be carefully selected for size, shape, and surface chemistry to optimize performance. Additionally, reducing or replacing toxic additives (like 6PPD and zinc oxide) is crucial to lower environmental and health risks. Testing with dynamic mechanical analysis helps fine-tune the balance between rolling resistance, grip, and wear.

In essence, it's about smart material selection, compound design, and minimizing harmful additives to achieve durable, safe, and environmentally friendly tyres.